

HHS3190M
HUMAN PHYSIOLOGY AND FUNCTIONAL ANATOMY
FOR REHABILITATION SERVICES

HHS3190MJ
復康服務之人體生理學及功能性解剖學

Physiology L6 Respiratory System

生理學 L6: 呼吸系統

(Sem 1 AY26-27 26-27學年第1學期)

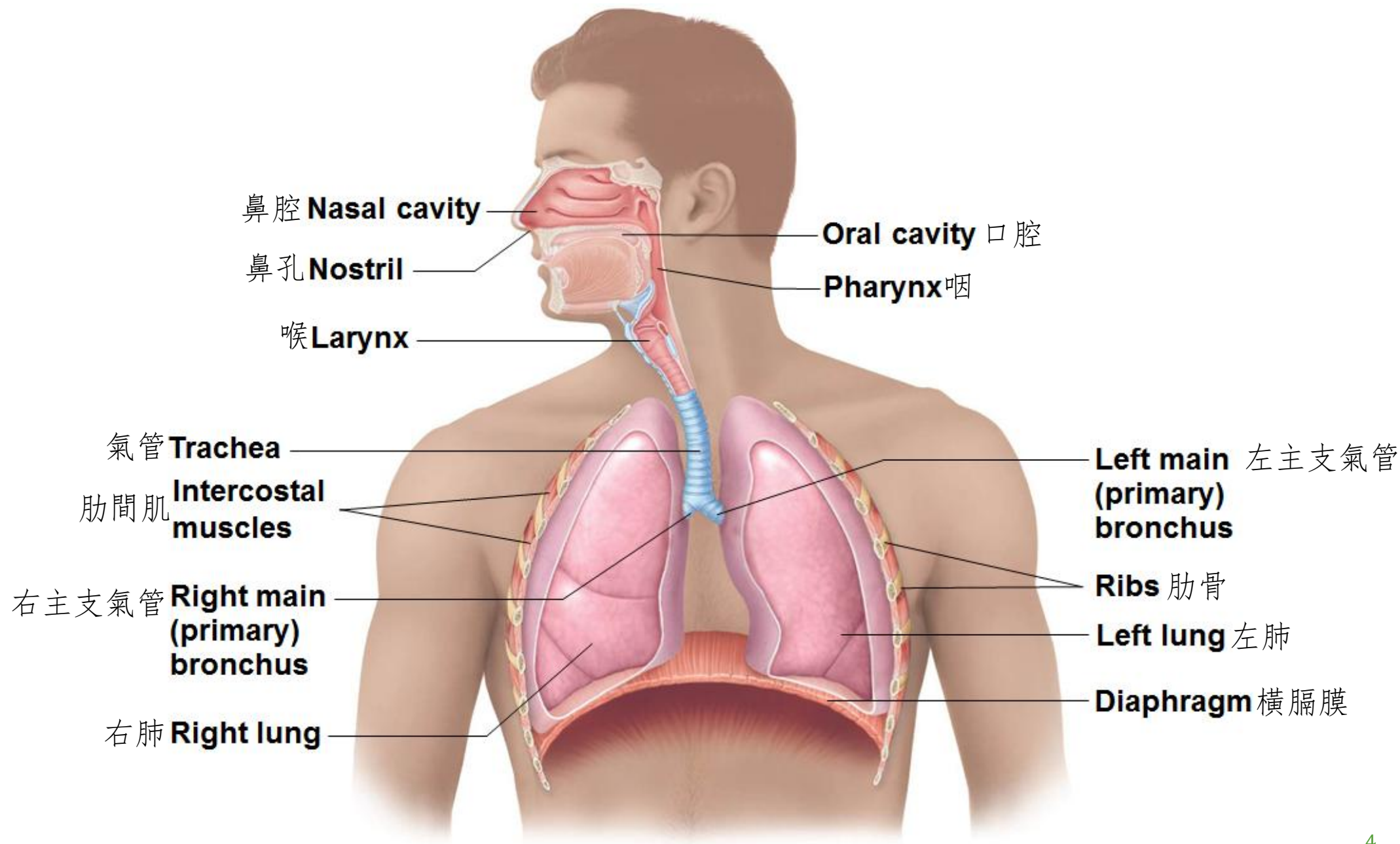
Outline大綱

- **Anatomy of the Respiratory System** 呼吸系統的解剖
 - Major Organs and Structures 主要器官和結構
 - Conducting Zone 傳導區
 - Respiratory Zone 呼吸區
- **Mechanics of Breathing** 呼吸力學
 - Inspiration and Expiration 吸氣與呼氣
 - Respiratory Volumes and Pulmonary Function Tests 呼吸量和肺功能測試
- **Gas Exchange between Lungs and Blood** 肺和血液之間的氣體交換
- **Transport of Gases in Blood** 血液中氣體的運輸
- **Control of Breathing** 呼吸控制

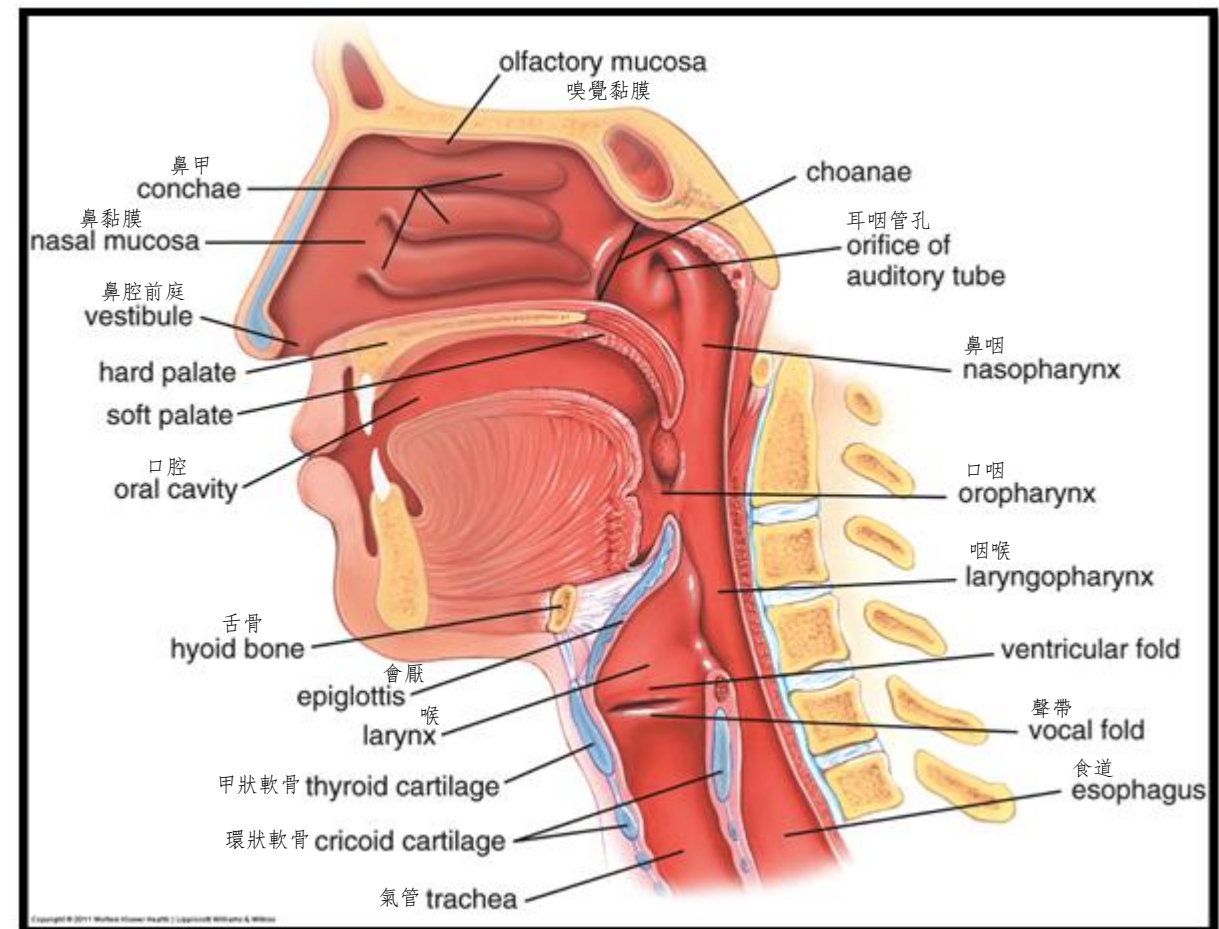
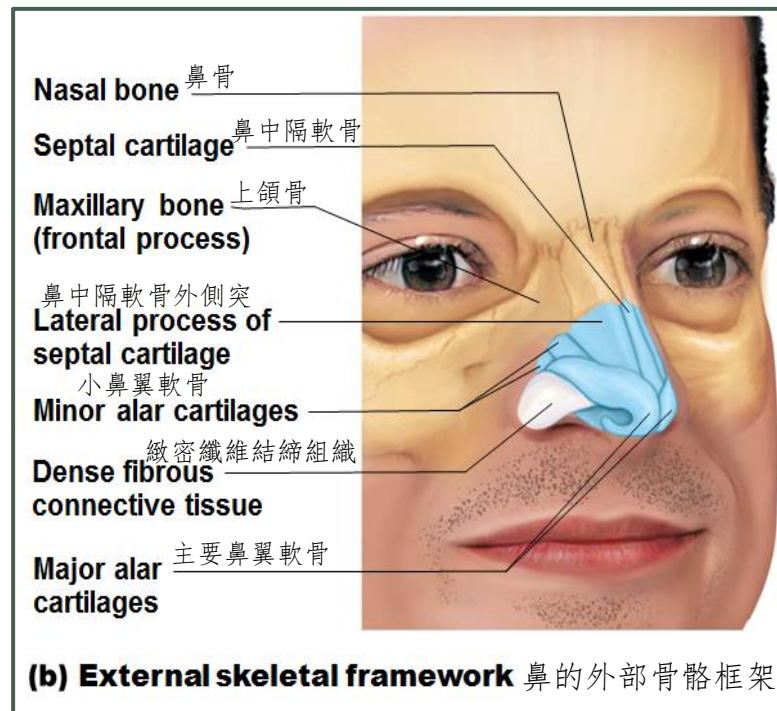
Anatomy of the Respiratory System

呼吸系統的解剖

- Major organs 主要器官
 - **Upper respiratory tract** 上呼吸道
 - Nose and nasal cavity 鼻和鼻腔
 - Paranasal sinuses 鼻竇
 - Pharynx 咽
 - Larynx 喉
 - **Lower respiratory tract** 下呼吸道
 - Trachea 氣管
 - Bronchi and branches (bronchiole) 支氣管及其分支(細支氣管)
 - Lungs and alveoli 肺和肺泡



Organ /Structure 器官/結構	Description 描述	Function 功能
Nose 鼻 (Nasal cavity 鼻腔)	- External part supported by bones and cartilages; internal cavity divided by nasal septum and lined with mucus. 鼻外部由骨骼和軟骨支撐;內腔被鼻中隔隔開，佈滿黏液。 - Roof of nasal cavity contains olfactory epithelium. 鼻腔頂部含有嗅覺上皮。	➤ Produce mucus for filtering and moistening air. 產生粘液以過濾和濕潤空氣。 ➤ Warming air 溫暖空氣 ➤ Sensation of smell 嗅覺
Pharynx 咽	- Passageway connecting (1) nasal cavity to larynx and (2) oral cavity to esophagus 連接 (1) 鼻腔與喉部 和 (2) 口腔與食道 的通道 - Have tonsils (lymphatic tissues) 有扁桃體 (淋巴組織)	➤ Passageway for food and air 食物和空氣之通道 ➤ Protection against inhaled pathogens 抵禦吸入的病原
Larynx 喉	- Connecting pharynx to trachea 連接咽及氣管 - Cover by cartilage and dense connective tissue 被軟骨和緻密的結締組織覆蓋 - Opening can be closed by epiglottis and vocal cords 開口可以通過會厭和聲帶閉合	➤ Passageway for air 空氣之通道 ➤ Epiglottis prevents food from entering trachea 會厭防止食物進入氣管 ➤ Voice production 發出聲音



Organ /Structure 器官/結構	Description 描述	Function 功能
Trachea 氣管	- A flexible tube begins just under the larynx and runs down behind the sternum 從喉部下方開始之具彈性管道，在胸骨後面向下延伸 - Wall contains C-shaped cartilages and smooth muscles in the posterior side 氣管壁包含C形軟骨和後側的平滑肌 - Line with ciliated epithelium with many mucus secreting cells 被有許多黏液分泌細胞的纖毛上皮覆蓋	➤ Passageway for air 空氣之通道 ➤ Clean, warm, and moisten inhaled air 清潔、溫暖、濕潤吸入的空氣
Bronchi 支氣管 (Single: bronchus)	- Consist of left and right main bronchi, then divide into smaller bronchi and bronchioles 由左右主支氣管組成，然後分為較小的支氣管和細支氣管 - Wall contains smooth muscles 支氣管壁具有平滑肌	➤ Passageway connecting trachea and alveoli in lungs 連接氣管和肺泡的通道 ➤ Clean, warm, and moisten inhaled air 清潔、溫暖、濕潤吸入的空氣 ➤ Constriction of bronchial tree impedes expiration while dilation enhances expiration 支氣管樹收縮阻礙呼氣，擴張則促進呼氣

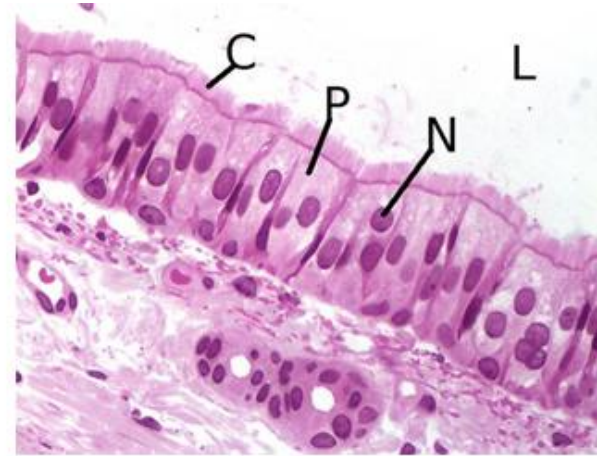
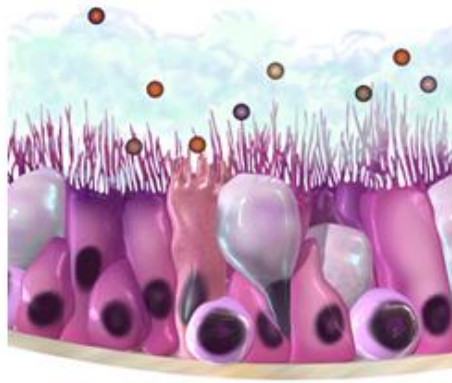
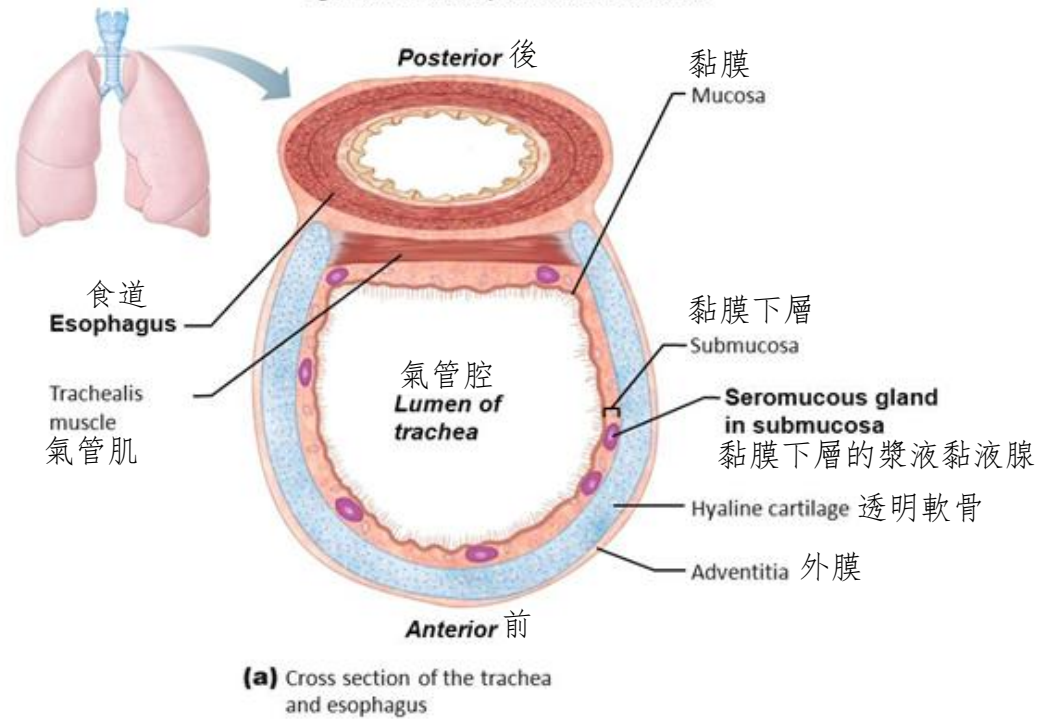
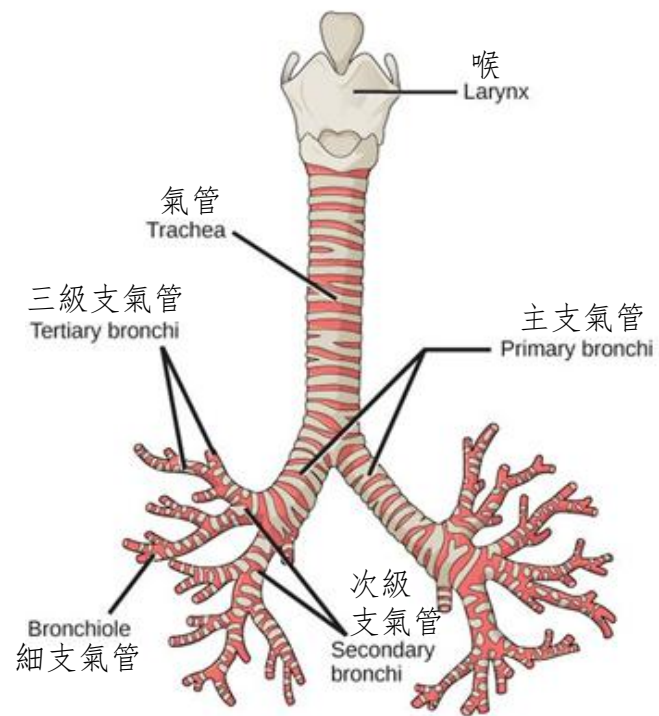
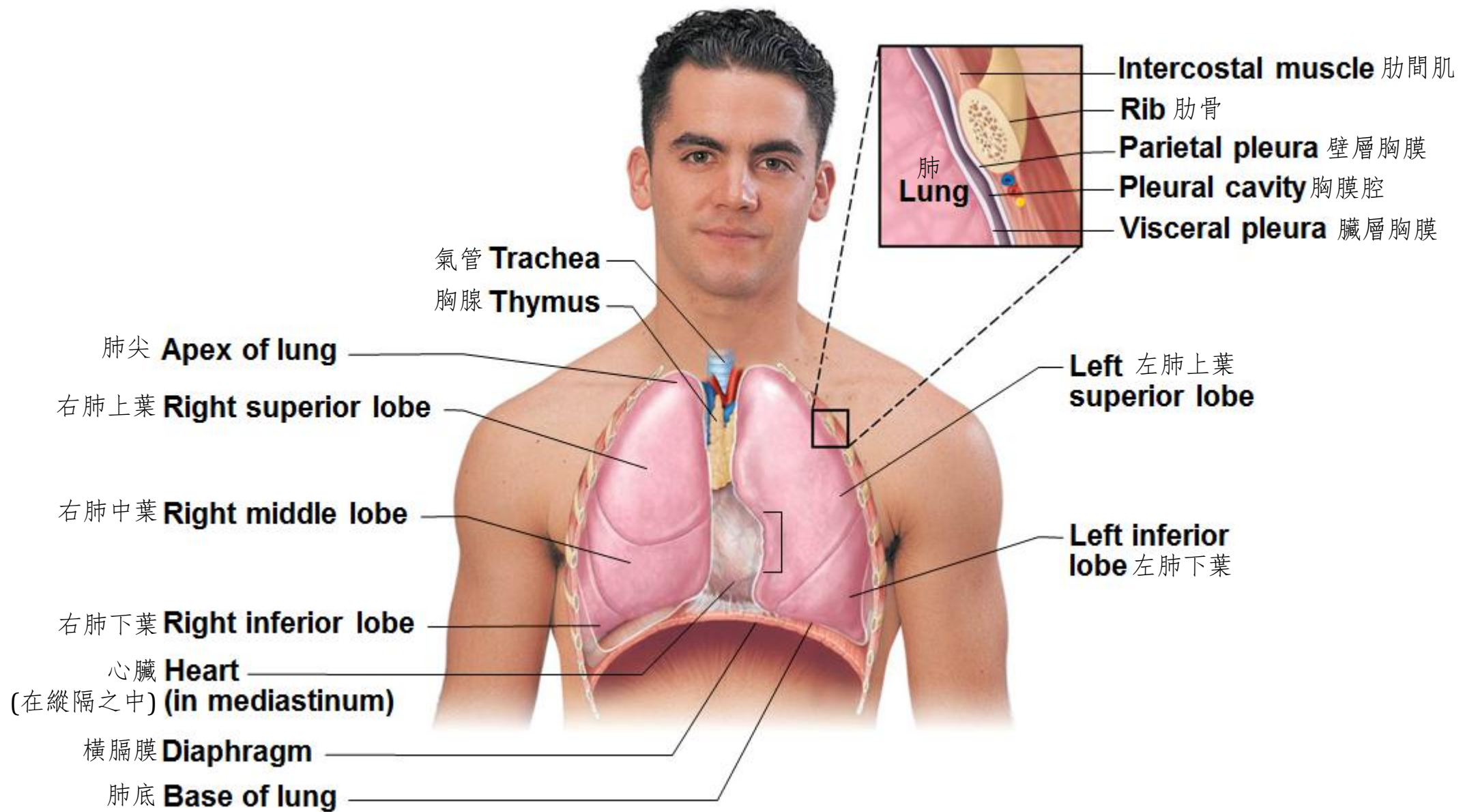
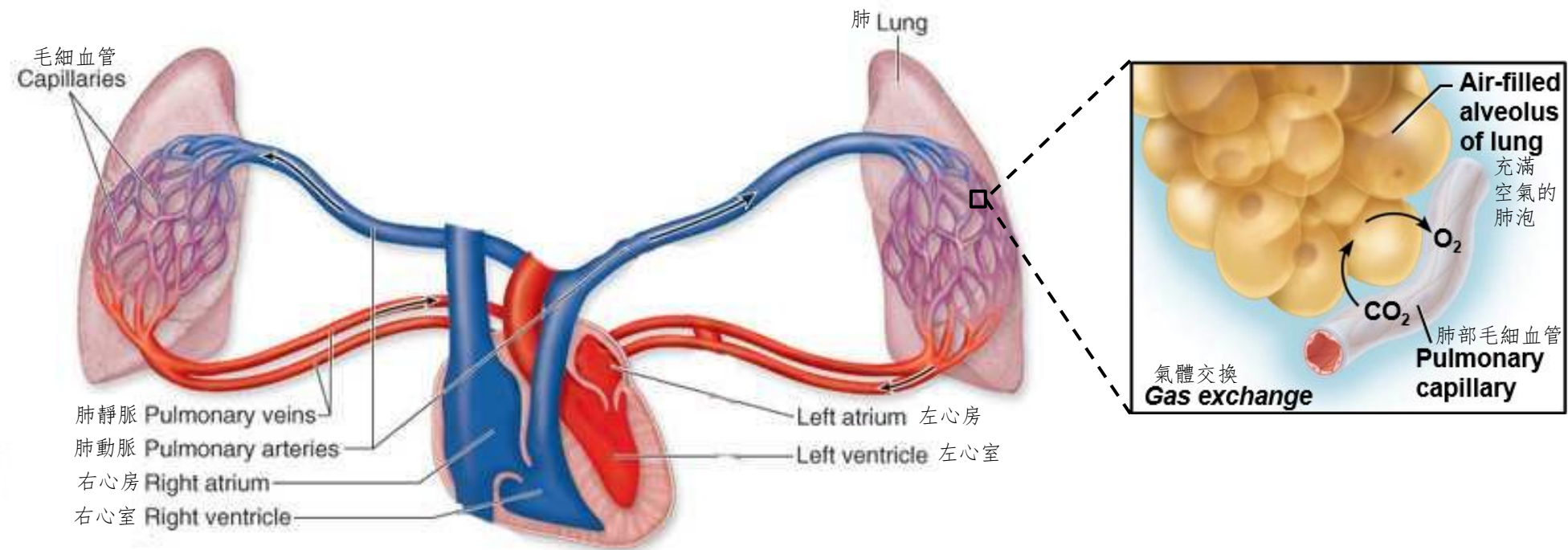


Figure 21.6a Tissue composition of the tracheal wall.



Organ /Structure 器官/結構	Description 描述	Function 功能
Lungs 肺	- Right and left lungs are located within the thoracic cavity of the chest 左右肺位於胸腔內 - Composed of alveoli, bronchial tree, and elastic connective tissues 由肺泡、支氣管樹和彈性結締組織組成	➤ Ventilation (Inhale and exhale) 通氣 (吸氣和呼氣) ➤ Gas exchange between air and blood 空氣和血液之間的氣體交換
Alveoli 肺泡 (Single: Alveolus)	- Smallest chamber at the ends of bronchial tree 支氣管樹末端最小的腔室 - Walls are simple squamous epithelium 肺泡壁是簡單鱗狀上皮 - External surfaces are surrounded by many capillaries 外表面被許多毛細血管包圍	➤ Site of gas exchange between inhaled air and blood 吸入空氣與血液進行氣體交換的部位
Pleura 胸膜	2 layers of serous membrane 2層漿膜 Parietal pleura lines thoracic cavity 壁層胸膜覆蓋胸腔 Visceral pleura covers external surfaces of lungs 臟層胸膜覆蓋肺部外表面	➤ Produce pleural fluid for lubrication during inhaling and exhaling movements 產生胸膜液，在吸氣和呼氣時作潤滑之用 ➤ Compartmentalize the lungs 劃分肺部





Connection between lungs and heart - Pulmonary circulation:

肺和心臟之間的連接 - 肺循環:

- **Right heart**右心 → **Pulmonary arteries**肺動脈 → **Lungs (Gas exchange)** 肺 (氣體交換)
- **Lungs**肺 → **Pulmonary veins**肺靜脈 → **Left heart**左心

Anatomy of the Respiratory System

呼吸系統的解剖

- The respiratory system is divided into 2 zones: 呼吸系統分為2個區域:

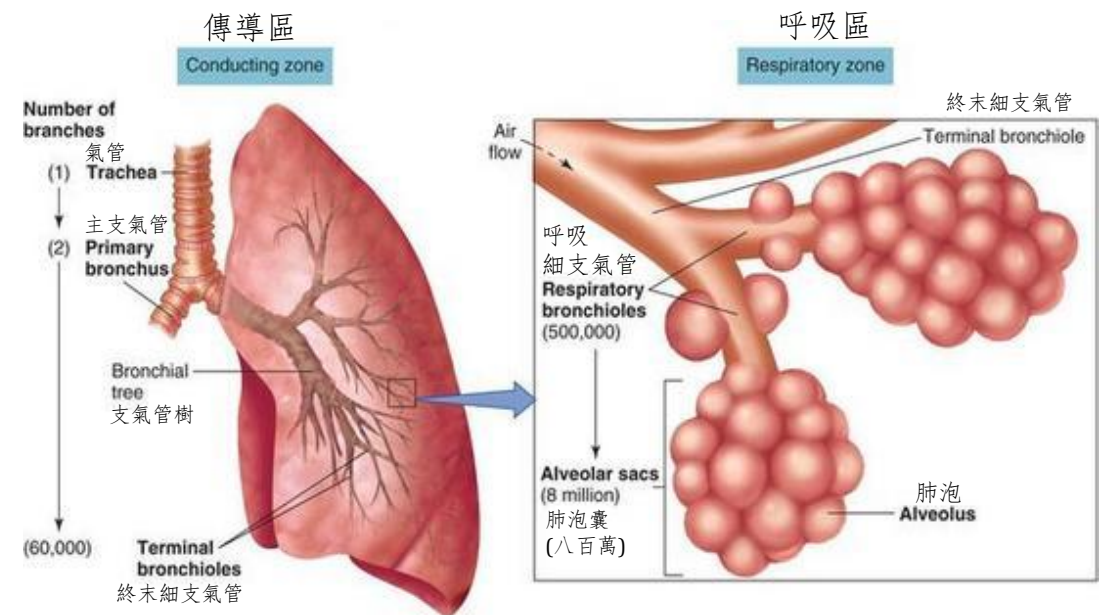
- **Conducting zone 傳導區**

- All passageways for air 所有空氣通道
- Allow **low resistance to air flow** 低氣流阻力
- **Filter, warm, and humidify air** before entering alveoli
在進入肺泡之前過濾、溫暖和濕潤空氣

- **Voice production** 發出聲音

- **Respiratory zone 呼吸區**

- **Respiratory bronchioles** (smallest bronchiole) and **alveoli**
呼吸細支氣管 (最小的細支氣管) 和 肺泡
- Actual site of **gas exchange** 氣體交換實際位置

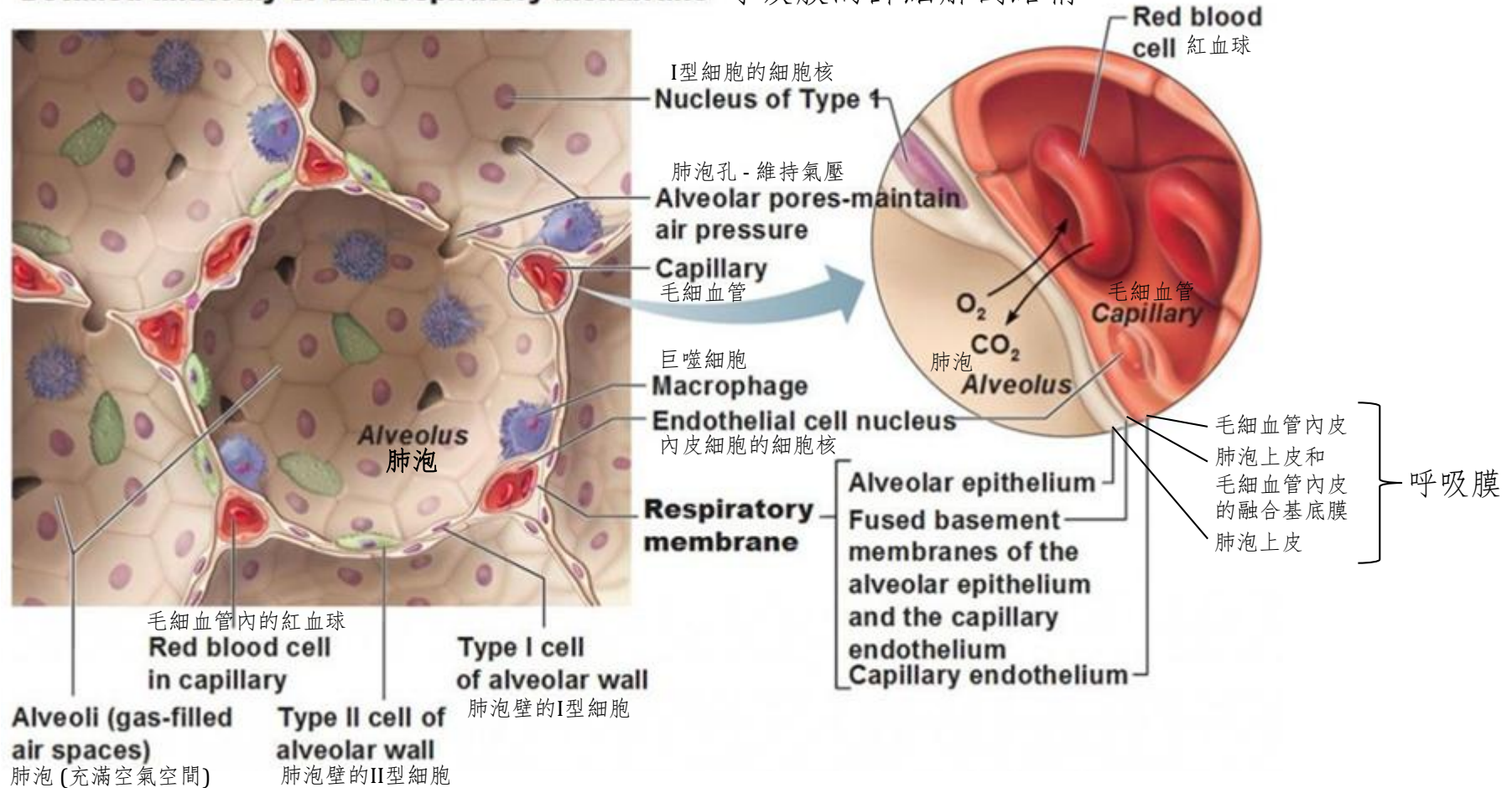


Anatomy of the Respiratory System

呼吸系統的解剖

- Respiratory zone structure 呼吸區結構
 - The **alveolar walls** are composed of 1 layer of squamous epithelial cells, called “type I cells”. **肺泡壁**由1層鱗狀上皮細胞組成，稱為「I型細胞」。
 - The external surface of type I cells is covered by a thin **basement membrane**.
I型細胞的外表面被一層薄的**基底膜**覆蓋。
 - A dense network of thin-wall **capillaries** surrounds the alveolus.
密集的薄壁**毛細血管**網絡圍繞著肺泡。
- Together, the alveolar wall, basement membrane, and the capillary wall form the **respiratory membrane**.
肺泡壁、基底膜和毛細血管壁共同形成**呼吸膜**。
- Gas exchange occurs by simple diffusion across the respiratory membrane.
氣體交換通過穿過呼吸膜的簡單擴散而發生。

Detailed anatomy of the respiratory membrane 呼吸膜的詳細解剖結構



Cells, called “**type II cells**”, are found on the alveolar walls. 在肺泡壁上有稱為 “**II型細胞**” 的細胞。

- These cells secrete a fluid containing **surfactant** to the internal surfaces of alveoli.
這些細胞向肺泡內表面分泌含有 **表面活性劑** 的液體。
- This helps to reduce surface tension and **prevent collapse of alveoli** during expiration.
這有助於降低表面張力並 **防止** 呼氣期間 **肺泡塌陷**。

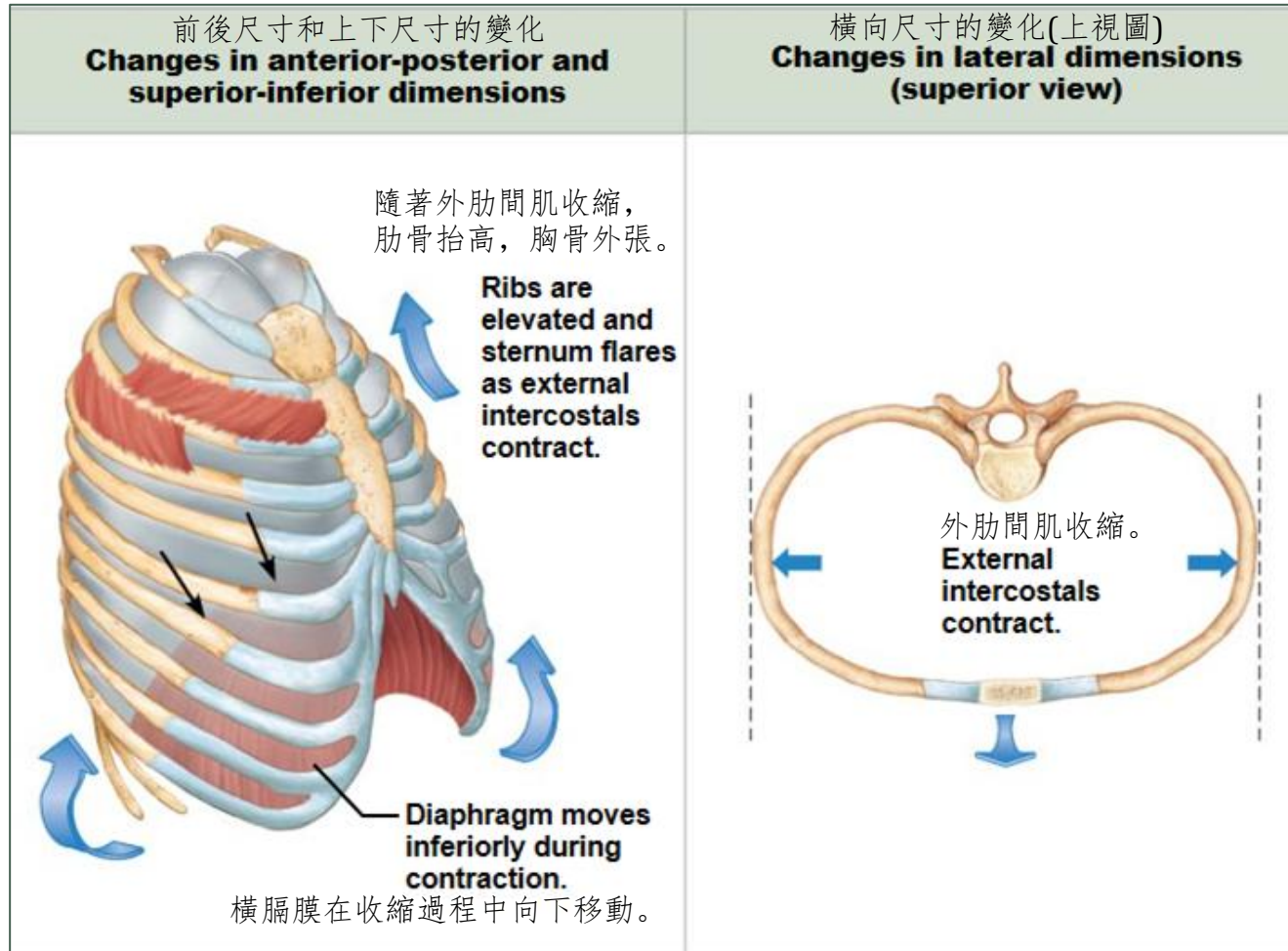
Mechanics of Breathing 呼吸力學

- Pulmonary ventilation - “Breathing” 肺通氣 - “呼吸”
 - consists of two phases 由兩個階段組成
 - Inspiration: gases flow into lungs 吸氣：氣體流入肺部
 - Expiration: gases exit lungs 呼氣：氣體從肺部排出
- Important basic sequence: 重要的基本順序：
 - (1) Muscle movements 肌肉運動
 - (2) Volume changes 容積變化
 - (3) Pressure changes 壓力變化
 - (4) Flows of air 空氣流動

- **Inspiratory muscles 吸氣肌 :**
 - **Diaphragm & intercostal muscles**
橫膈膜和肋間肌
 - Contract to induce inspiration
收縮以引發吸氣
 - Relax to induce expiration
放鬆以引發呼氣
- **Thoracic cavity 胸腔 :**
 - **Chest cavity 胸腔**
 - The chamber that encloses by chest wall (ribs, sternum and intercostal muscles) and diaphragm
由胸壁 (肋骨、胸骨和肋間肌) 和橫膈膜包圍的腔室
 - Include pleural cavity, lungs, heart, and other organs
包括胸膜腔、肺、心臟等器官

- **Intrapulmonary volume 肺內容積 :**
 - **Lung volume 肺容量**
 - **Alveolar volume 肺泡容積**
- **Intrapulmonary pressure (P_{alv}) 肺內壓 :**
 - **Pressure in the alveoli 肺泡內壓力**
 - When it is higher than P_{atm} , air flows out of lungs
當高於 P_{atm} 時，空氣會從肺部流出
 - When it is lower than P_{atm} , air flows into lungs
當低於 P_{atm} 時，空氣流入肺部
- **Atmospheric pressure (P_{atm}) 大氣壓力 :**
 - At sea level, is ~ 760 mmHg (=1 atm)
在海平面為 ~ 760 毫米汞柱 (=1 atm)
 - Count as **Zero** in respiration process
呼吸過程中算作零

Inspiration 吸氣



Inspiratory muscles contract
(diaphragm descends; rib cage rises).
吸氣肌收縮 (橫膈膜下降; 胸腔上升)。



Thoracic cavity volume increases.
胸腔容積增加。



Lungs are stretched.
Intrapulmonary volume increases.
肺部被拉伸，肺內容積增加。

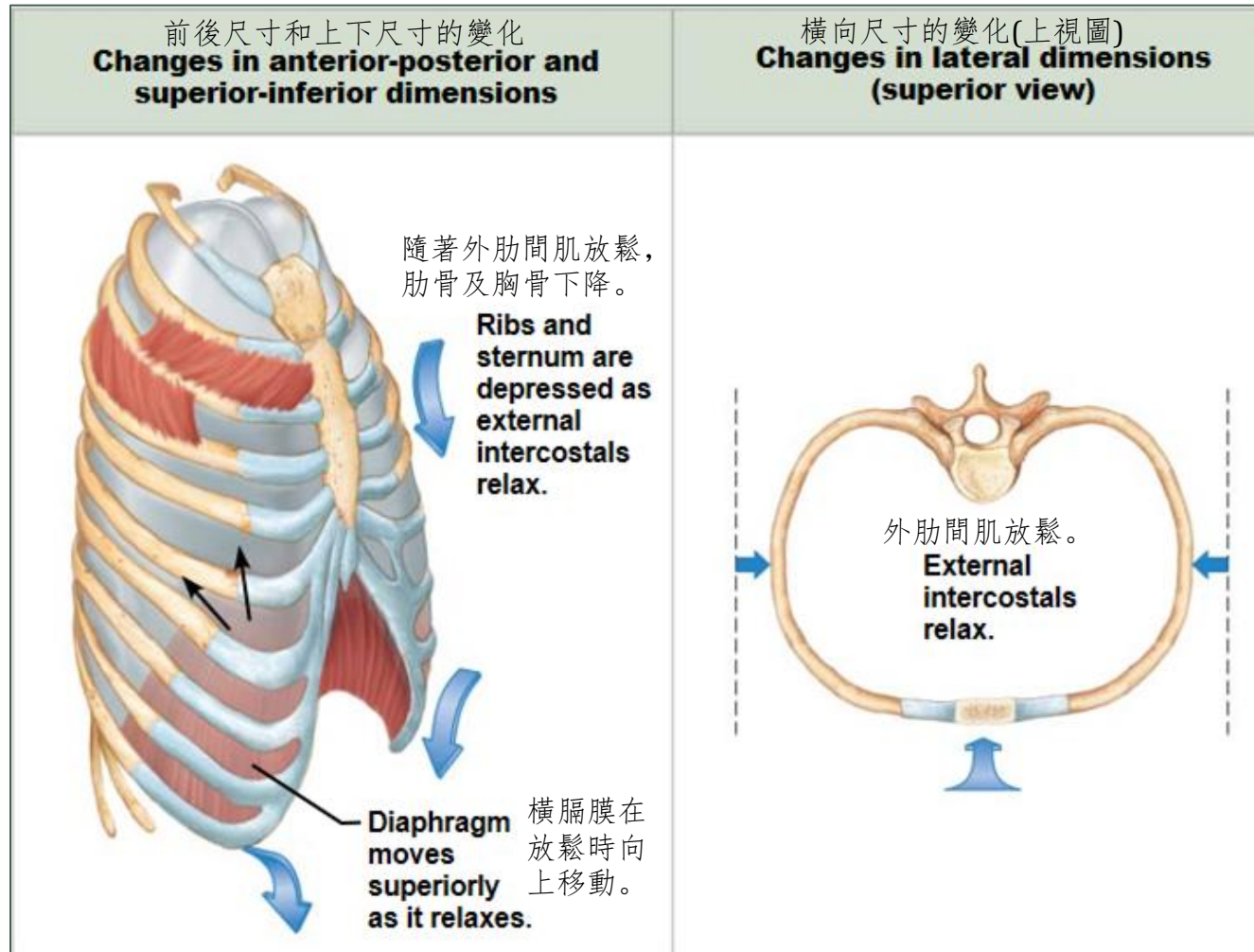


Intrapulmonary pressure
drops to -1 mmHg.
肺內壓降至-1 mmHg。



Air flows into lungs down the pressure
gradient until intrapulmonary pressure is 0.
空氣沿著壓力梯度流入肺部，
直到肺內壓為 0。

Expiration 呼氣



Inspiratory muscles relax
(diaphragm rises; rib cage descends due to recoil of costal cartilages).
吸氣肌放鬆(橫膈膜上升；肋骨因肋軟骨回縮而下降)。

Thoracic cavity volume decreases.
胸腔容積減少。

Elastic lungs recoil passively.
Intrapulmonary volume decreases.
彈性肺部被動回縮，肺內容積減少。

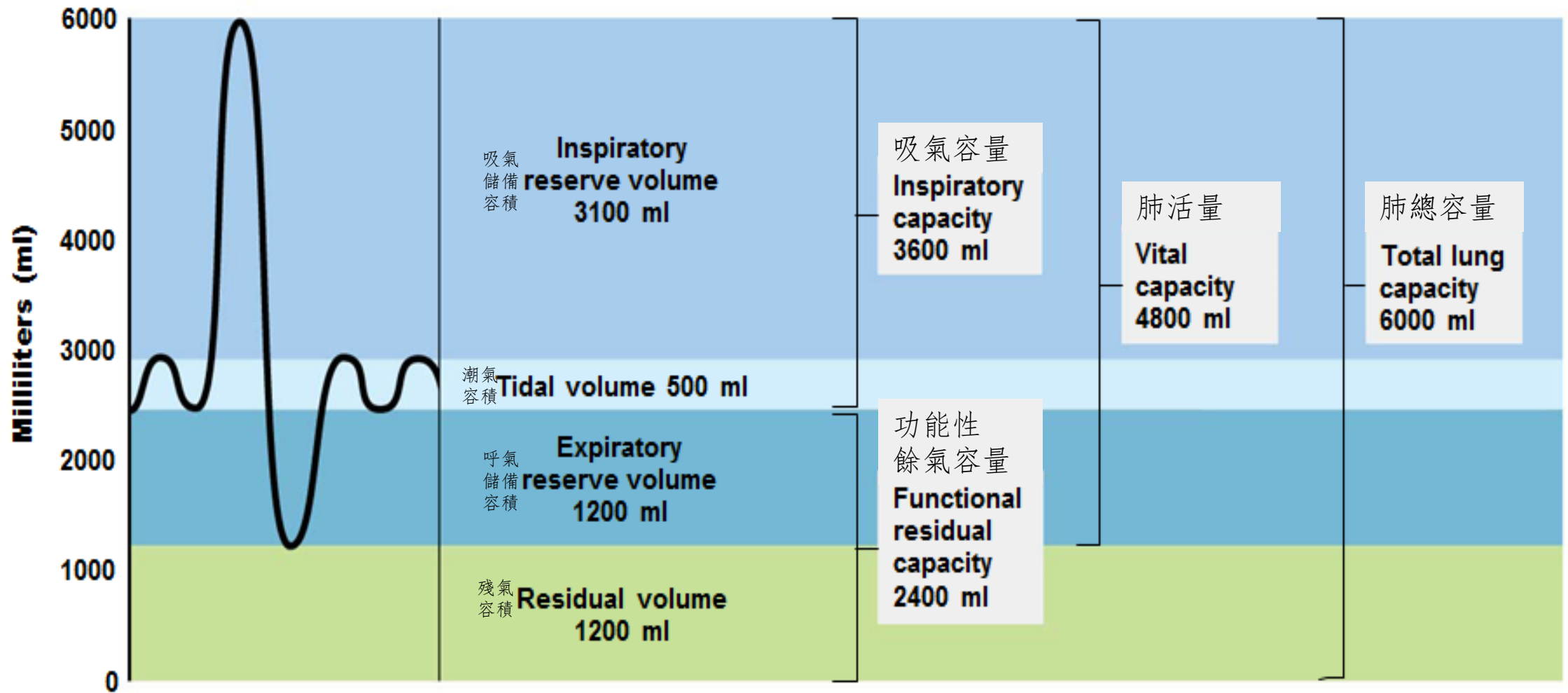
Intrapulmonary pressure
rises to +1 mmHg.
肺內壓升至+1 mmHg。

Air flows out of lungs down the pressure gradient
until intrapulmonary pressure is 0.
空氣沿著壓力梯度流出肺部，
直到肺內壓為 0。

Mechanics of Breathing 呼吸力學

- Several **respiratory volumes** can be used to **assess status of pulmonary ventilation** 可以使用多種呼吸容積來評估肺通氣狀況
- Respiratory volumes can be combined to calculate respiratory capacities, which can give information on a person's respiratory status
可以將呼吸容積結合起來計算呼吸容量，從而提供有關個人呼吸狀況的信息
- Respiratory volumes and capacities are **usually abnormal in people with pulmonary disorders** 肺部疾病患者的呼吸容積和呼吸容量通常異常
- **Spirometer**: original clinical tool used to measure patient's respiratory volumes
肺活量計：用於測量患者呼吸容積的原始臨床工具
- Electronic measuring devices used today
當今使用電子測量設備

Name 名稱		Average Value (Adult) 平均值 (成人)		Description 描述
		Male 男性	Female 女性	
Respiratory volume 呼吸容積	Tidal Volume (TV) 潮氣容積	500 ml	500 ml	Amount of air inhaled or exhaled with each breath under resting conditions 靜息條件下每次呼吸吸入或呼出的空氣量
	Inspiratory Reserve Volume (IRV) 吸氣儲備容積	3100 ml	1900 ml	Amount of air that can be forcefully inhaled after a normal tidal volume inspiration 在正常潮氣容積吸氣後可以盡力額外吸入的空氣量
	Expiratory Reserve Volume (ERV) 呼氣儲備容積	1200 ml	700 ml	Amount of air that can be forcefully exhaled after a normal tidal volume expiration 在正常潮氣容積呼氣後可以盡力額外呼出的空氣量
	Residual Volume (RV) 殘氣容積	1200 ml	1100 ml	Amount of air remaining in the lungs after a forced expiration 盡力呼氣後肺部剩餘的空氣量
Respiratory Capacity 呼吸容量	Total Lung Capacity (TLC) 肺總容量	6000 ml	4200 ml	Maximum amount of air contains in lung after a maximum inspiration. 最大吸氣後肺部含有的最大空氣量。 $TLC = TV + IRV + ERV + RV$
	Vital Capacity (VC) 肺活量	4800 ml	3100 ml	Maximum amount of air that can be exhaled after a maximum inspiration. 最大吸氣後可以呼出的最大空氣量。 $VC = TV + IRV + ERV$
	Inspiratory Capacity (IC) 吸氣容量	3600 ml	2400 ml	Maximum amount of air that can be inhaled after a normal tidal volume expiration. 在正常潮氣容積呼氣後可以吸入的最大空氣量。 $IC = TV + IRV$
	Functional Residual Capacity (FRC) 功能性餘氣容量	2400 ml	1800 ml	Volume of air remaining in the lungs after a normal tidal volume expiration. 在正常潮氣容積呼氣後肺部剩餘的空氣量。 $FRC = ERV + RV$



(a) Spirographic record for a male 男性肺活量圖

Mechanics of Breathing 呼吸力學

- **Anatomical dead space** 解剖死腔 :
 - Volume of air that **does not take part in gas exchange** 不參與氣體交換的空氣量
 - Consists of air that remains in **conducting zone** 由留在傳導區的空氣組成
 - **Average 150 ml** out of 500 ml TV 在潮氣容積為500 ml之情況下，平均值為**150 ml**
- **Alveolar dead space** 肺泡死腔 :
 - Should be **zero** in healthy person 在健康人仕身上應該為**零**
 - Space occupied by nonfunctional alveoli 非功能性肺泡佔據的空間
 - Can be due to collapse or obstruction 可能由於塌陷或阻塞
- **Total dead space** 死腔總容量 :
 - **sum of anatomical and alveolar dead space** 解剖死腔及肺泡死腔之總和

Mechanics of Breathing 呼吸力學

- **Pulmonary Function Test 肺功能測試**
- Spirometry can distinguish 肺活量測定可以區分
 - **Obstructive pulmonary disease:** increased airway resistance (example: bronchitis)
阻塞性肺病：氣道阻力增加 (例如：支氣管炎)
 - **TLC, FRC, RV may increase**
TLC、FRC、RV 可能會增加
 - **Restrictive disease:** reduced TLC due to disease (example: tuberculosis) or exposure to environmental agents (example: fibrosis)
限制性肺病：由於疾病(例如：肺結核)或暴露於環境因素(例如：纖維化)而導致 TLC 減少
 - **VC, TLC, FRC, RV decrease** because lung expansion is compromised
因為肺擴張受限，VC、TLC、FRC、RV 減少

Mechanics of Breathing 呼吸力學

- **Pulmonary Function Test 肺功能測試**

- **Forced vital capacity (FVC) 用力肺活量：**

- amount of gas forcibly expelled after taking deep breath 深呼吸後盡力呼出的氣體量

- **Forced expiratory volume (FEV) 用力呼氣容積：**

- amount of gas expelled during specific time interval of FVC
FVC 特定時間間隔內呼出的氣體量

- **FEV₁**: amount of air expelled in 1st second
1秒用力呼氣容積：第一秒排出的空氣量

- Healthy individuals can expel **80% of FVC in 1st second**
健康人仕可以在第一秒內呼出**FVC的80%**

- Patients with obstructive disease exhale less than 80% in 1st second
阻塞性疾病的患者在第一秒呼氣量不足80%

- Patients with restrictive disease exhale 80% or more even with reduced FVC
限制性疾病的患者呼氣量仍可達 80% 或更多，同時FVC降低

Mechanics of Breathing 呼吸力學

- **Minute Ventilation (MV) 每分鐘通氣量：**
 - total amount of gas that flows into or out of *respiratory tract* in 1 minute
1 分鐘內流入或流出呼吸道的氣體總量
 - Normal at rest: Average 6 L/min
靜止時正常值：平均 6 公升/分鐘
 - Normal with exercise: Up to 200 L/min
運動時正常值：高達 200 公升/分鐘
 - Only rough estimate of respiratory efficiency 僅粗略估計呼吸效率
- **$MV = TV \times \text{Respiratory Rate}$ 每分鐘通氣量 = 潮氣容積 $\times$ 呼吸頻率**
- Example: TV = 500 ml; Respiratory Rate = 14 breaths/min
例子：潮氣容積 = 500 毫升; 呼吸頻率 = 14 次呼吸/分鐘

$$MV = 500 \times 14 = 7000 \text{ ml/min}$$
$$\text{每分鐘通氣量} = 500 \times 14 = 7000 \text{ 毫升/分鐘}$$

Mechanics of Breathing 呼吸力學

- **Alveolar ventilation (AV) 肺泡通氣量：**
 - Flow of gases into and out of *respiratory zone* in 1 minute
1分鐘內流入和流出呼吸區的氣體量
 - Volume of dead space is taken into account in AV 肺泡通氣量考慮了死腔的容積
 - **Better indicator of effective ventilation 有效通氣的更好指標**
- **$AV = (TV - \text{Dead space}) \times \text{Respiratory Rate}$**
肺泡通氣量 = (潮氣容積 - 死腔容積) $\times$ 呼吸頻率
- Example: TV = 500 ml; Respiratory Rate = 14 breaths/min; Dead space = 150 ml
例子：潮氣容積 = 500 毫升; 呼吸頻率 = 14 次呼吸/分鐘; 死腔 = 150 毫升

$$AV = (500 - 150) \times 14 = 4900 \text{ ml/min}$$
$$\text{肺泡通氣量} = (500 - 150) \times 14 = 4900 \text{ 毫升/分鐘}$$

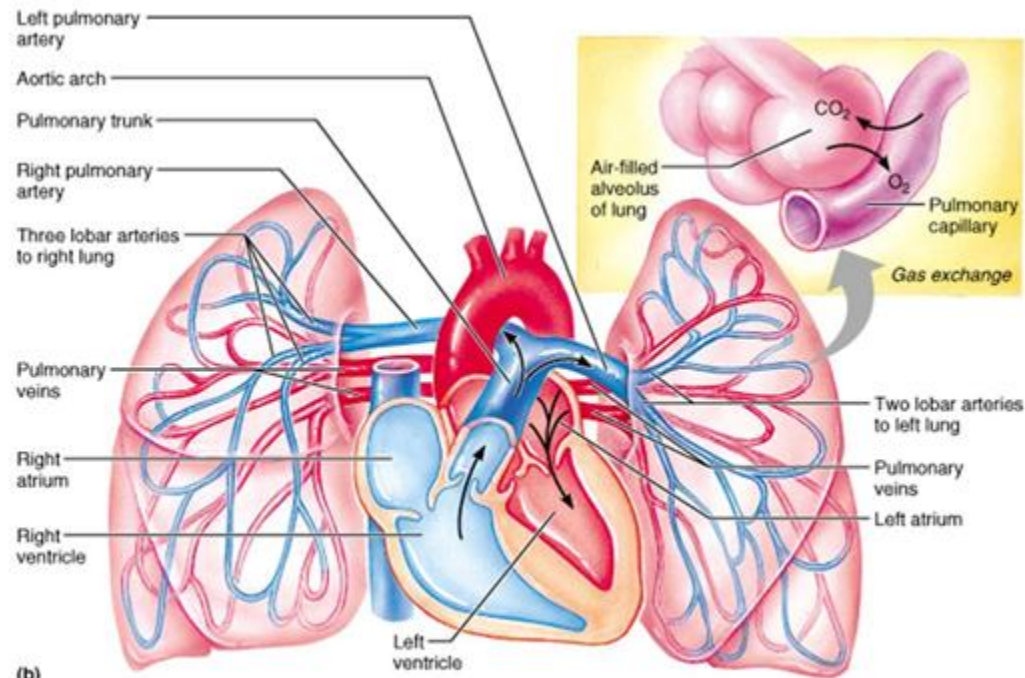
Mechanics of Breathing 呼吸力學

- Because dead space in an individual is normally constant, AV is mainly affected by TV and frequency 由於個體的死腔通常是恆定的，因此 AV 主要受 TV 和呼吸頻率的影響
- Significant increases in AV are brought about by increasing TV rather than frequency AV 的顯著增加是透過增加TV而不是呼吸頻率來實現的
- Rapid, shallow breathing can actually decrease AV 快速而淺的呼吸實際上會降低AV

Breathing Pattern 呼吸模式	TV (ml) 潮氣容積	Dead space (ml) 死腔容積	Respiratory Rate (breaths/min) 呼吸頻率	MV (ml/min) 每分鐘通氣量	AV (ml/min) 肺泡通氣量	% Effective Ventilation 有效通氣百分比 [AV/MV]
Normal rate and depth 正常速率和深度	500	150	20			
Slow and deep breathing 緩慢而深的呼吸	1000	150	10			
Rapid and shallow breathing 急促而淺的呼吸	200	150	50			

Gas Exchange between Lungs and Blood

肺和血液之間的氣體交換



- “**Blue**” in this figure 這張圖中的“藍色”部份:
- **Deoxygenated blood** containing less O₂ and more CO₂ 缺氧血含有較少的氧和較多的二氧化碳
- “**Red**” in this figure 這張圖中的“紅色”部份:
- **Oxygenated blood** containing more O₂ and less CO₂ 含氧血含有較多的氧和較少的二氧化碳
- **Gaseous exchange in lungs** 肺部氣體交換:
- **O₂ up-taking:** O₂ diffuse from alveoli to blood 氧吸收: 氧從肺泡擴散到血液
- **CO₂ unloading:** CO₂ diffuse from blood to alveoli 二氧化碳卸載: 二氧化碳從血液擴散到肺泡

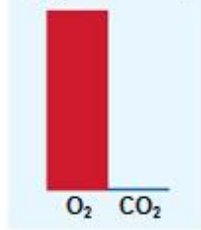
Gas Exchange between Lungs and Blood

肺和血液之間的氣體交換

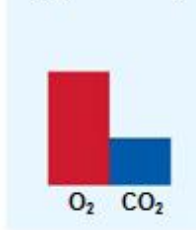
- **Partial pressures and gas concentrations** 分壓和氣體濃度
 - (1) The pressure exerted by each gas (Partial Pressure) in a mixture of gases is directly proportional to the percentage of the gas in the mixture.
氣體混合物中每種氣體施加的壓力(分壓)與混合物中氣體的百分比成正比。
 - E.g. 21% O₂ & 760 mmHg P_{O₂} = 160 mmHg
例如 21% 氧氣 和 760 mmHg 氧氣分壓 = 160 mmHg
 - (2) When a mixture of gases dissolves in liquid, the concentration of each gas in the liquid will be directly proportional to its partial pressure.
當氣體混合物溶解在液體中時，液體中每種氣體的濃度將與其分壓成正比。
 - i.e. **higher partial pressure ⇒ higher concentration** 即更高的分壓⇒更高的濃度
 - **partial pressure gradient ⇒ concentration gradient** 分壓梯度⇒濃度梯度
- When gas diffuse across the respiratory membrane 當氣體擴散穿過呼吸膜
 - ***Region of higher partial pressure → Region of lower partial pressure***
分壓較高的區域 → 分壓較低的區域
 - Down the partial pressure gradient!!! 沿著分壓梯度向下!!

大氣空氣：
氧氣分壓 160 mmHg
二氧化碳分壓 0.3 mmHg

Atmospheric air:
 P_{O_2} 160 mm Hg
 P_{CO_2} 0.3 mm Hg



Air in alveoli:
 P_{O_2} 104 mm Hg
 P_{CO_2} 40 mm Hg

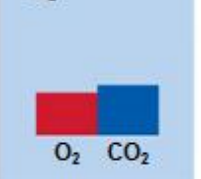


肺泡空氣：
氧氣分壓 104 mmHg
二氧化碳分壓 40 mmHg

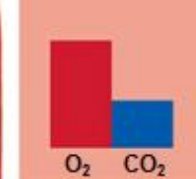
Gas exchange
between lungs
and blood
肺和血液之間的
氣體交換

進入肺部之血液：
氧氣分壓 40 mmHg
二氧化碳分壓 45 mmHg

Blood entering
lungs:
 P_{O_2} 40 mm Hg
 P_{CO_2} 45 mm Hg



Blood leaving
lungs:
 P_{O_2} 100 mm Hg
 P_{CO_2} 40 mm Hg



離開肺部之血液：
氧氣分壓 100 mmHg
二氧化碳分壓 40 mmHg

Gas exchange
between blood
and tissue
血液和組織之間的
氣體交換

肺動脈
Pulmonary
arteries

肺靜脈
Pulmonary
veins

Alveoli
肺泡

Heart

Systemic veins
體循環
靜脈

Systemic arteries
體循環
動脈

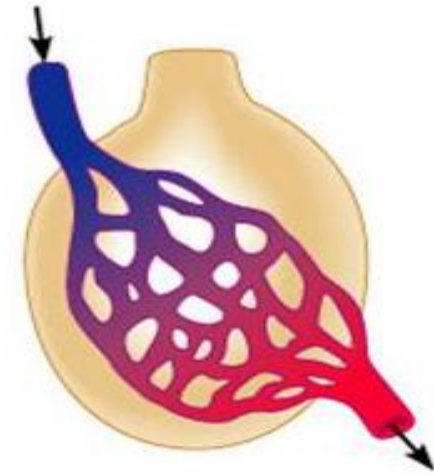
Tissues:
 P_{O_2} less than 40 mm Hg
 P_{CO_2} greater than 45 mm Hg

組織：
氧氣分壓 低於40 mmHg
二氧化碳分壓 高於45 mmHg

Gas Exchange between Lungs and Blood

肺和血液之間的氣體交換

- Other factors affecting gas exchange across respiratory membrane
影響呼吸膜氣體交換的其他因素
- Matching of ventilation and blood flow in pulmonary capillaries
肺毛細血管血流與通氣的相互配合
- Thickness and surface area of respiratory membrane
呼吸膜的厚度和表面積



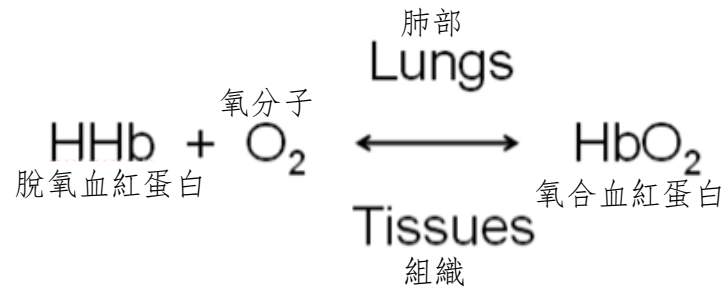
Transport of Gases in Blood 血液中氣體的運輸

- **Oxygen Transport 氧氣運輸**

- Molecular O₂ is carried in blood in two ways:
氧分子以兩種方式在血液中運輸：
- (1) 1.5% is **dissolved in plasma** 1.5%溶解在血漿中
- (2) 98.5% is loosely **bound to each Fe atom of hemoglobin (Hb) in RBCs**
98.5%鬆散地與紅細胞中血紅蛋白(Hb)的每個鐵原子結合
- Each Hb molecule is composed of four polypeptide chains, each with 1 iron-containing heme group
每個 Hb 分子由四條多肽鏈組成，每條多肽鏈有 1 個含鐵血紅素基團
- So each Hb can transport four oxygen molecules
所以每個 Hb 可以運輸四個氧分子
- **Oxyhemoglobin (HbO₂)**: hemoglobin-O₂ combination
氧合血紅蛋白 (HbO₂): 血紅素-氧分子結合
- **Deoxyhemoglobin (HHb)**: hemoglobin that has released O₂
脫氧血紅蛋白 (HHb): 已釋放出氧分子的血紅蛋白

Transport of Gases in Blood 血液中氣體的運輸

- In **lungs**, the deoxyhemoglobin molecules (HHb) uptake O_2 and become oxyhemoglobin (HbO_2)
在**肺部**，脫氧血紅蛋白分子(HHb)吸收氧分子(O_2)並成為氧合血紅蛋白(HbO_2)
- **$HHb + O_2 \rightarrow HbO_2$**
- In **tissues**, the HbO_2 dissociates again and the O_2 diffuses into tissue cells for metabolic reactions
在**組織**中，氧合血紅蛋白(HbO_2)再次解離，氧分子(O_2)擴散到組織細胞中進行代謝反應
- **$HbO_2 \rightarrow HHb + O_2$**

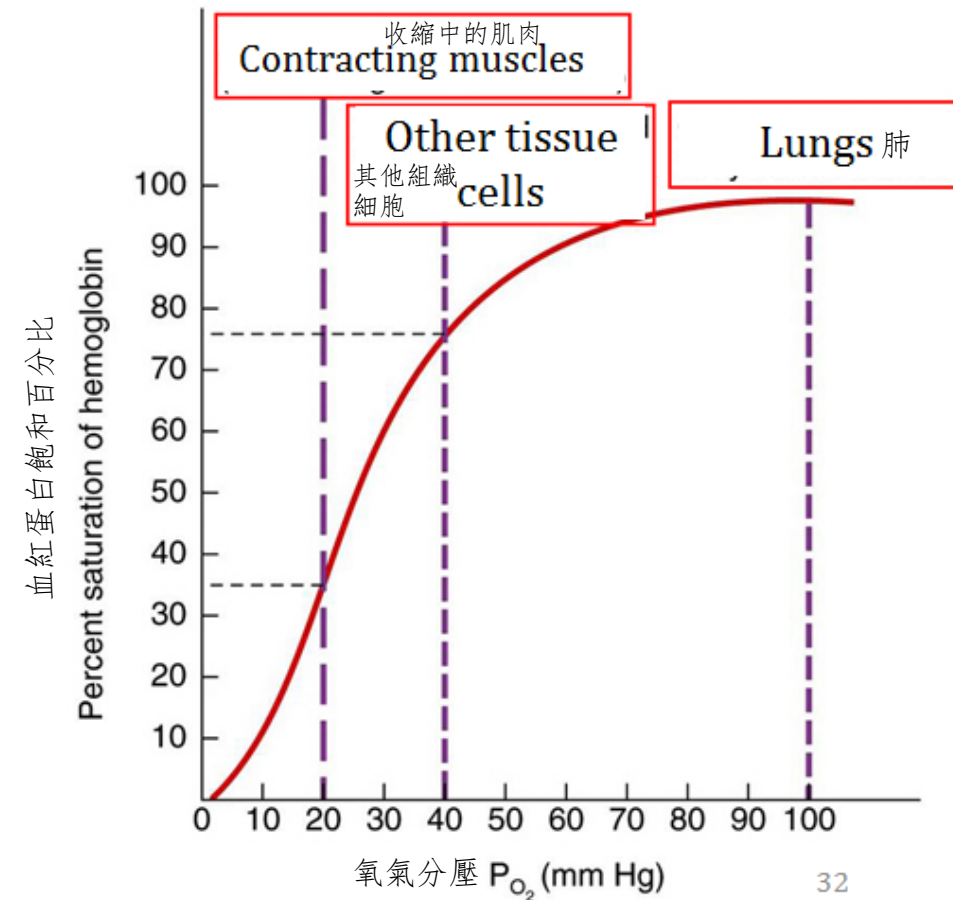


Transport of Gases in Blood 血液中氣體的運輸

- **Hemoglobin Saturation**

- 血紅蛋白飽和度

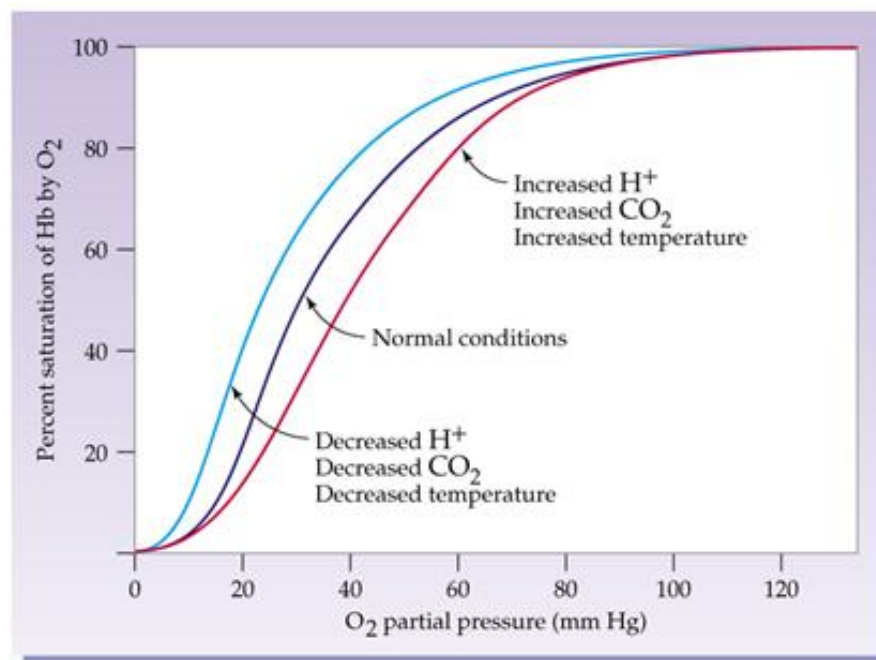
- When all heme groups of hemoglobin are bound to O_2 , the hemoglobin is saturated.
當血紅蛋白的所有血紅素基團都已經與氧分子結合時，血紅蛋白就飽和。
 - The level of hemoglobin saturation depends on PO_2 of the body part!
血紅蛋白飽和度的水平取決於身體部位的氧氣分壓！



Copyright © John Wiley & Sons, Inc. All rights reserved.

Transport of Gases in Blood 血液中氣體的運輸

- **More O₂ released** when:
更多氧分子在以下情況下被釋放:
 - **Higher temperature** E.g. fever/ heavy exercise 較高溫度 例如發燒/劇烈運動
 - **Blood becomes more acidic** E.g. heavy exercise 血液變得更酸性 例如劇烈運動
 - **CO₂ level increases** E.g. heavy exercise 二氧化碳水平增加 例如劇烈運動

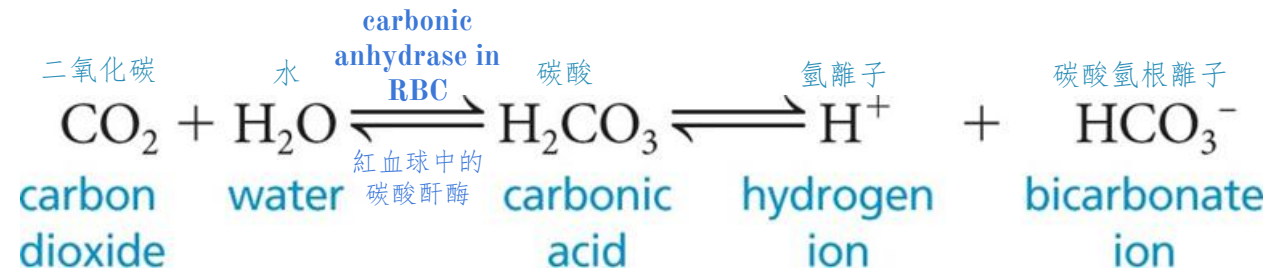


Transport of Gases in Blood 血液中氣體的運輸

- **Carbon Dioxide Transport 二氧化碳運輸**

- CO₂ is transported in blood in three ways:
二氧化碳通過三種方式在血液中運輸：

- (1) 10% is **dissolved in plasma** 10%溶解在血漿中
- (2) 20% is **bound to the globin part of hemoglobin and become carbaminohemoglobin (HbCO₂)**
20%與血紅蛋白的珠蛋白部分結合，成為氨基甲酸酯血紅蛋白(HbCO₂)
 - $\text{Hb} + \text{CO}_2 \rightarrow \text{HbCO}_2$
- (3) 70% is transported as **bicarbonate ions in plasma**
70%以碳酸氫根離子的形式在血漿中運輸
 - CO₂ **combining with water in red blood cells** to form **carbonic acid**, which quickly dissociates into **bicarbonate** and hydrogen ions in plasma
二氧化碳與紅血球中的水結合形成碳酸，碳酸在血漿中迅速解離成碳酸氫根和氫離子

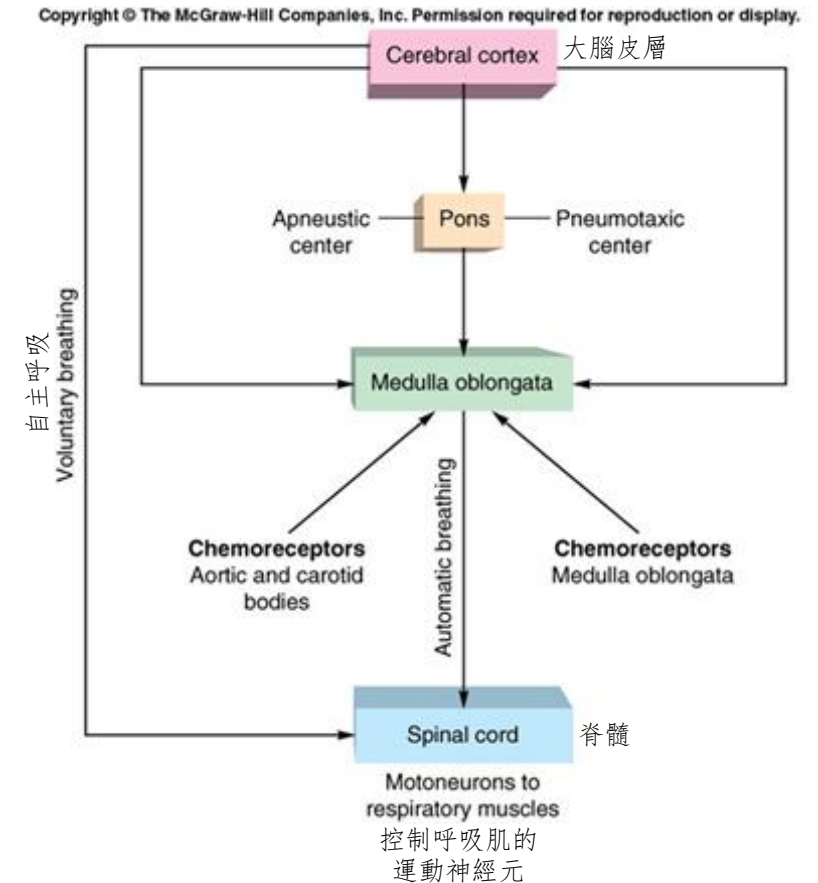


Transport of Gases in Blood 血液中氣體的運輸

- Influence of CO₂ on blood pH 二氧化碳對血液酸鹼值的影響
 - The carbonic acid-bicarbonate buffer system is very important to keep normal blood pH. 碳酸-碳酸氫鹽緩衝系統對於保持正常的血液酸鹼值非常重要。
 - Normal range of blood pH: **7.35-7.45** 血液酸鹼值正常範圍：**7.35-7.45**
- Change in respiratory rate can change blood pH by altering amount of carbonic acid in blood.
呼吸頻率的變化可以透過改變血液中的碳酸含量來改變血液的酸鹼值。
 - **Slow, shallow breathing**→More CO₂ remain in blood →More carbonic acid→**Blood pH decreases (Acidic)**
緩慢而淺的呼吸→更多的二氧化碳殘留在血液中→更多的碳酸→**血液酸鹼值降低(酸性)**
 - **Rapid, deep breathing**→Less CO₂ remain in blood →Less carbonic acid→**Blood pH increases (Alkaline)**
快速而深呼吸→血液中殘留的二氧化碳較少→碳酸較少→**血液酸鹼值升高(鹼性)**

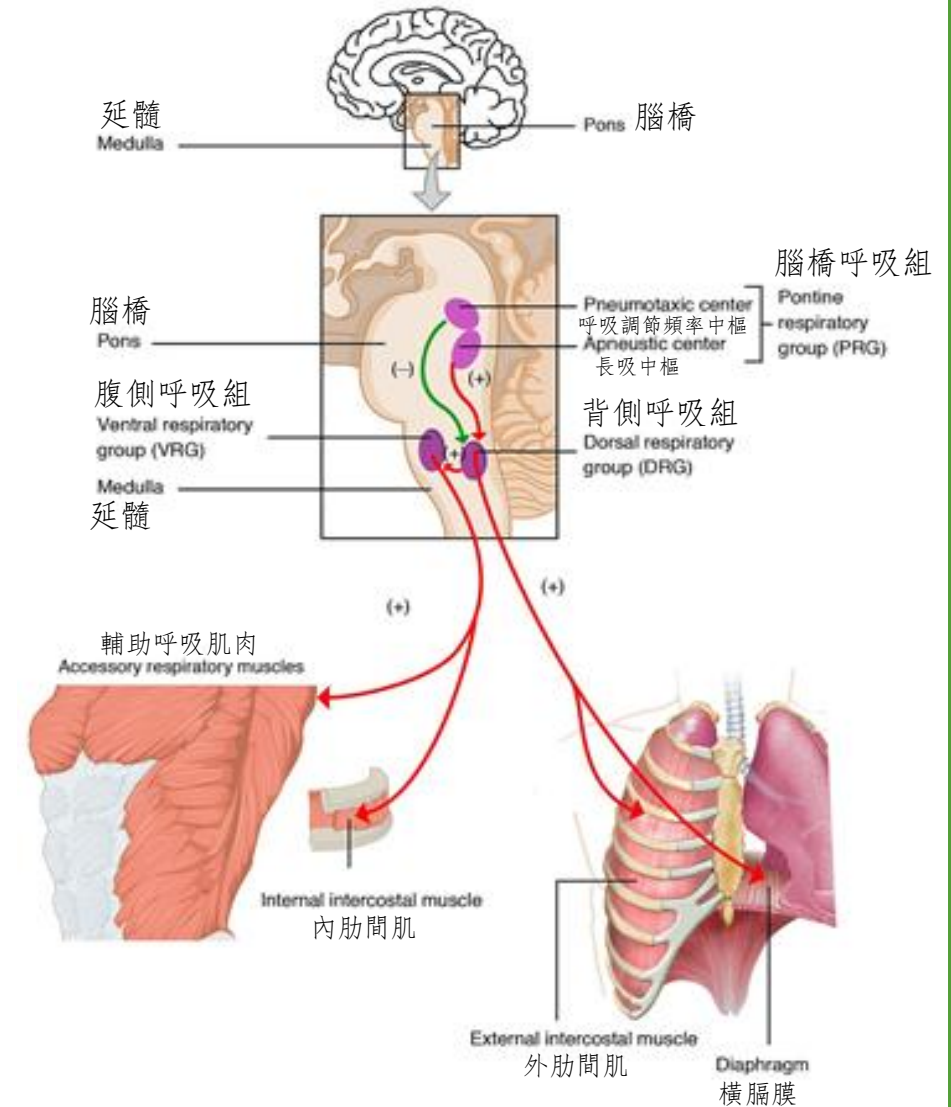
Control of Breathing 呼吸控制

- Respiratory rhythms are regulated by voluntary and involuntary control mechanisms
呼吸節律受自主和非自主控制機制調節
- Voluntary control of breathing** 自主呼吸控制
- Human can exert conscious control over the rate and depth of our breathing.
人類可以有意識地控制呼吸的速度和深度。
- The **cerebral motor cortex** send signals to motor neurons that **control the respiratory muscles**.
大腦運動皮層向控制呼吸肌的運動神經元發送信號。
- The respiratory centers in brainstem are bypassed.
腦幹的呼吸中樞被繞過。
- However, the ability of holding breathe voluntarily is limited.
然而，自主屏住呼吸的能力是有限的。



Control of Breathing 呼吸控制

- **Involuntary control of breathing** 非自主呼吸控制
- **Respiratory centers in brainstem** 腦幹的呼吸中樞
 - Motor neurons from respiratory centers in brainstem transmit neural signals to affect contraction of **respiratory muscles**.
來自腦幹呼吸中樞的運動神經元傳遞神經信號以影響呼吸肌的收縮。
 - The rate and depth of breathing are controlled.
呼吸的頻率和深度受到控制。



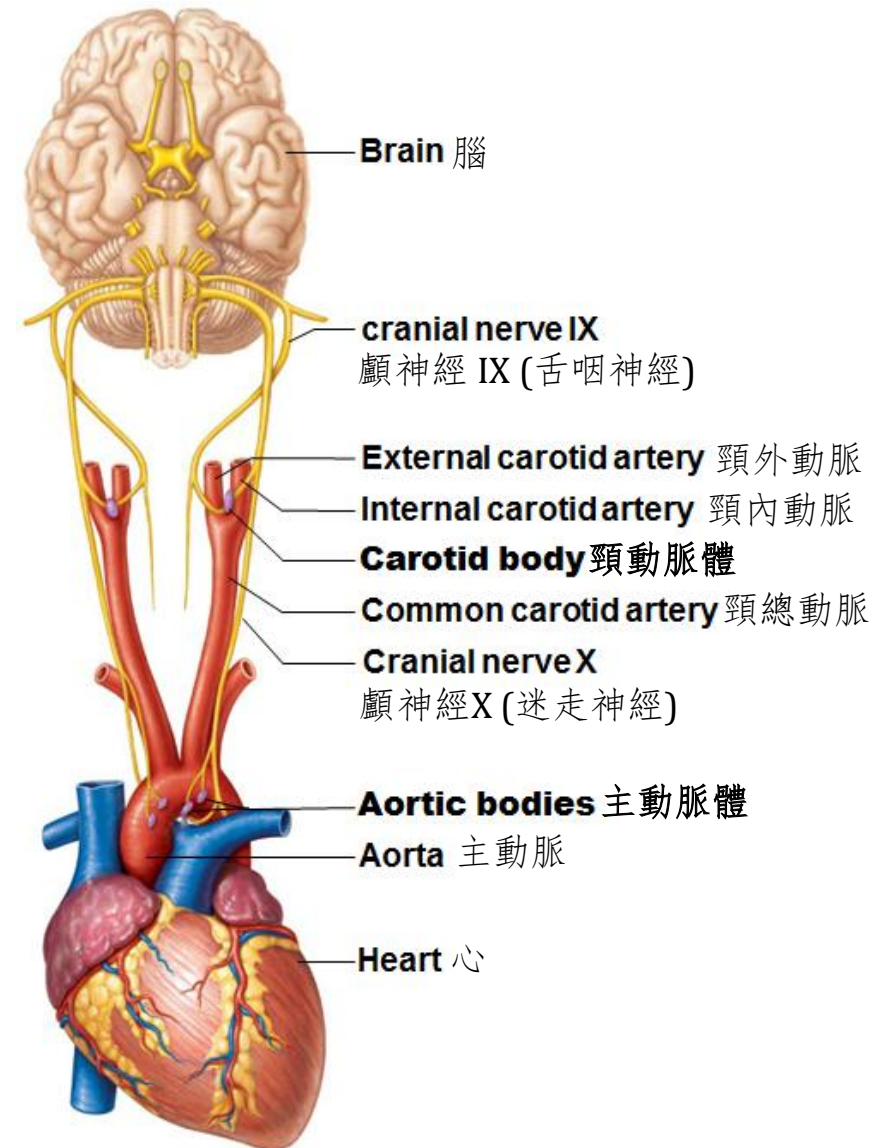
Control of Breathing 呼吸控制

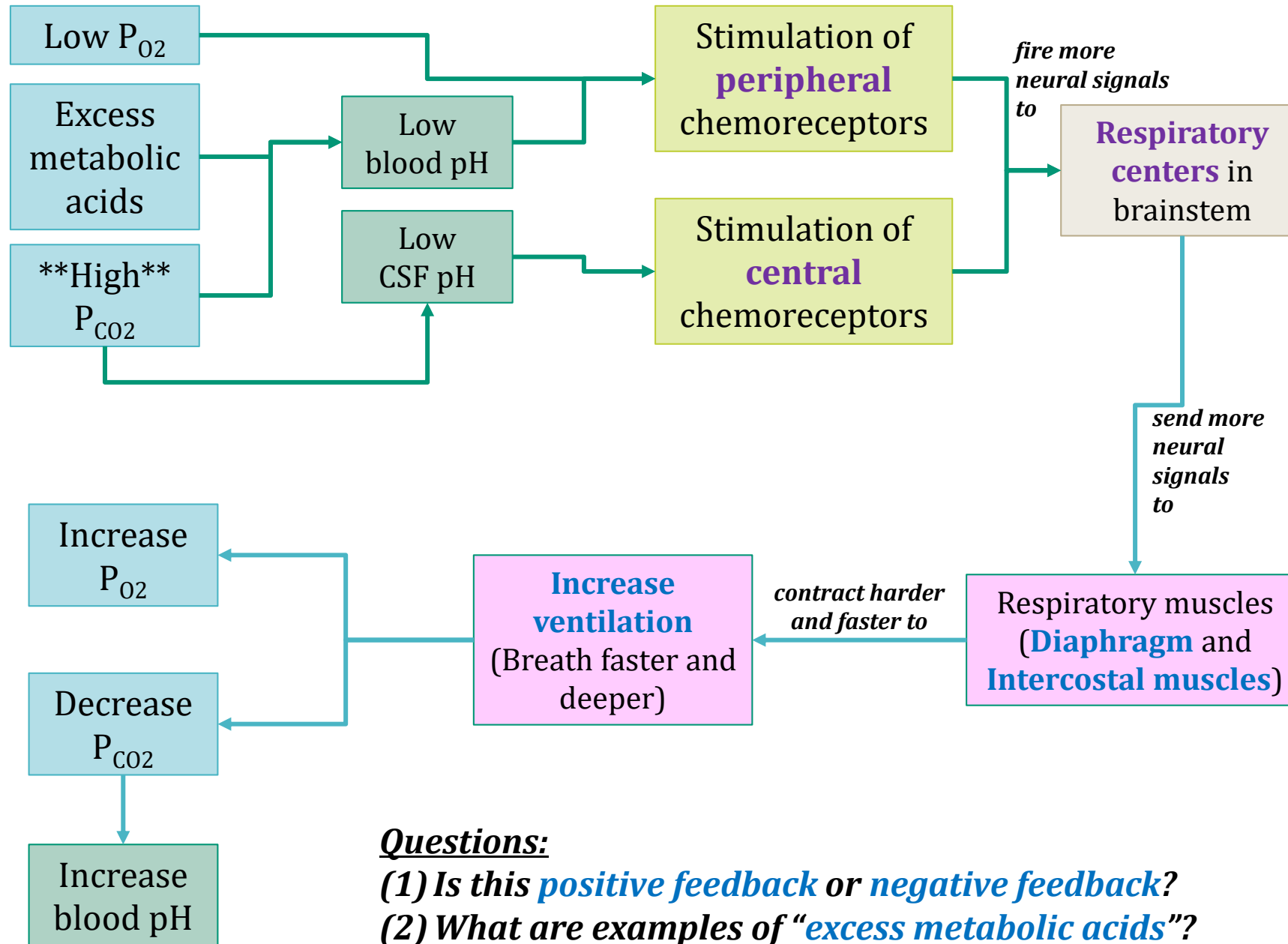
- **Involuntary control of breathing** 非自主呼吸控制
 - Changing levels of **carbon dioxide, oxygen, and hydrogen ions in arterial blood** will influence the respiratory centers and alter breathing pattern.
動脈血中二氧化碳、氧氣和氫離子水平的變化會影響呼吸中樞並改變呼吸模式。
 - **Chemoreceptors** 化學感受器：
 - detect changes in **blood PO_2 , PCO_2 and pH**
偵測血液氧氣分壓、二氧化碳分壓和酸鹼值的變化
 - (1) **Peripheral chemoreceptors** 外周化學感受器：
 - in carotid artery (**carotid bodies**) and around the aortic arch (**aortic bodies**)
在頸動脈(頸動脈體)和主動脈弓周圍(主動脈體)
 - (2) **Central chemoreceptors** 中樞化學感受器：
 - in **medulla oblongata** 在延髓

Control of Breathing

呼吸控制

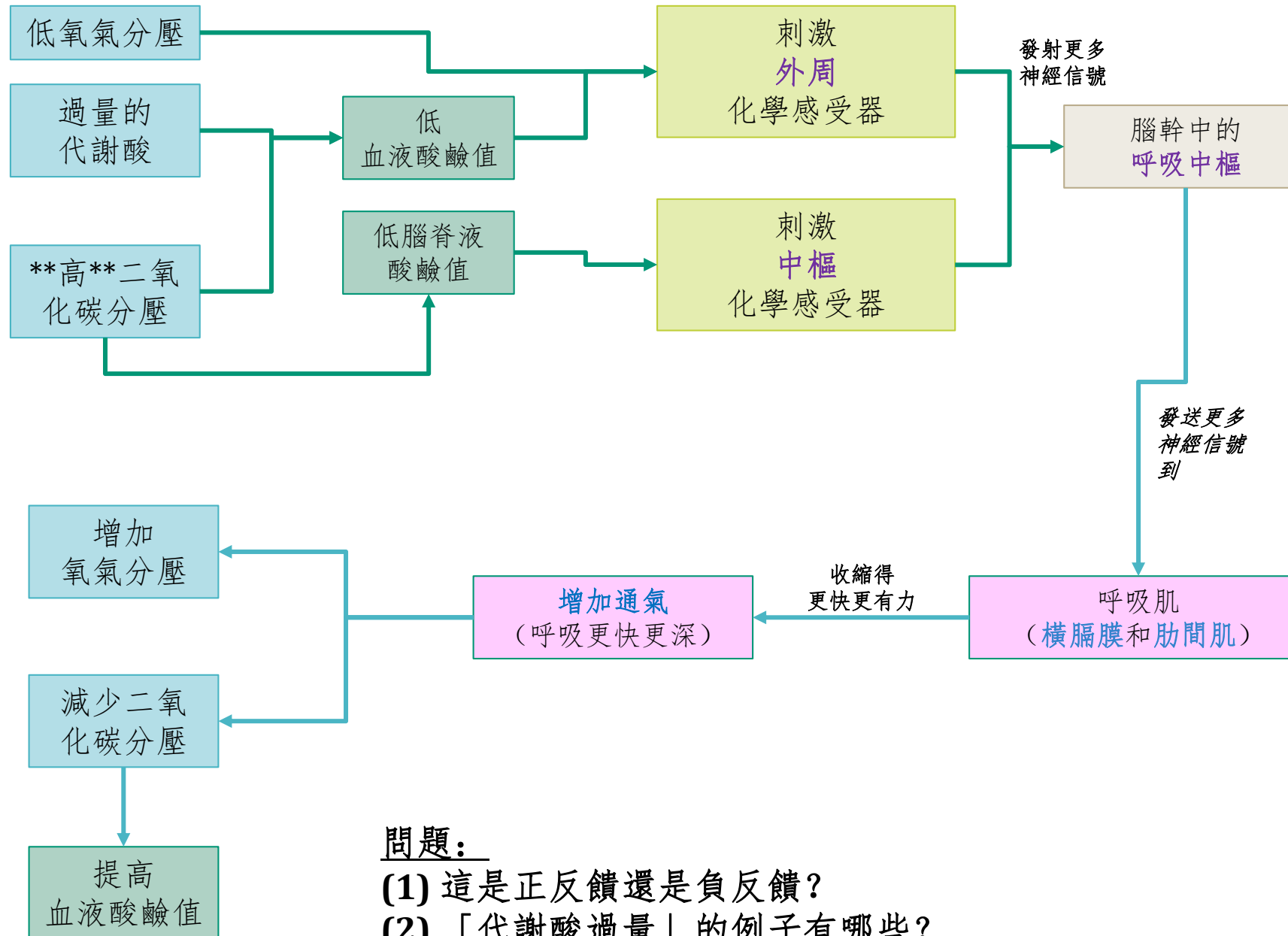
- **Involuntary control of breathing** 非自主呼吸控制
 - When these peripheral **chemoreceptors** detect **decreased levels of PO_2 , increased levels of PCO_2 , or decreased blood pH**
當這些外周化學感受器檢測到氧氣分壓水平降低、二氧化碳分壓水平升高或血液酸鹼值降低時
 - They fire more neural signals to **respiratory centers** in **brainstem**
它們向腦幹的呼吸中樞發射更多的神經信號
 - via sensory nerve fibers in **vagus nerve (cranial nerve X)** and **glossopharyngeal nerve (cranial nerve IX)**
通過迷走神經(顱神經X)和舌咽神經(顱神經IX)中的感覺神經纖維





Questions:

- (1) Is this **positive feedback** or **negative feedback**?
- (2) What are examples of “**excess metabolic acids**”?



問題：

- (1) 這是正反饋還是負反饋？
- (2) 「代謝酸過量」的例子有哪些？

L6 Revision Exercise 複習練習

Screen the QR code below to complete all questions in this revision exercise.
掃描下面的二維碼以完成此複習練習中的所有問題。



References

- 梅盛平, & 徐国昌. (2022). 正常人体结构学. 重庆大学电子音像出版社有限公司.
- 周华, & 杨向群 (2022). 人体解剖生理学(第8版). 人民卫生出版社.
- 管又飞, 罗自强, & 薛明明主编. 医学生理学(第5版). (2024). 北京大学医学出版社.
- 中华人民共和国国家卫生健康委员会. 常用临床医学名词. (2023).
<http://www.nhc.gov.cn/yzygj/s7659/202403/bdb6d9140b734323b49c9764deba8c45.shtml>
- Marieb, E.N., Hoehn, K. (2018) Human Anatomy & Physiology (11th ed.). Pearson International ed.
- Martini, F. H. (2017). Fundamentals of Anatomy and Physiology (11th ed.). England: Pearson Benjamin Cummings.
- Olson, T.R., Pawlina Woichiech (2015) A.D.A.M. Student Atlas of Anatomy (2nd ed.) Cambridge University Press. <https://doi.org/10.1017/CBO9780511800160>
- Snell, R.S. (2007) Clinical Anatomy by Systems. Lippincott Williams & Wilkins